Ain Shams University

Faculty of Engineering

CHEP – Communications Systems Engineering Program

Spring 2026 | ECE359: Digital Image Processing & Multimedia

Sheet 1

Digital Image Fundamentals & Interpolation

Part A Textbook Problems

1. [1] Using background knowledge of the human visual system and the structure of the fovea, and reasoning *purely in geometric terms*, estimate the diameter (in millimetres) of the smallest printed dot that the human eye can resolve when the page is held **0.2 m** away. Use the following assumptions:

- The visual system ceases to detect a dot once its image on the fovea becomes smaller than the diameter of a single cone receptor.
- The fovea is modelled as a square array of side **1.5 mm**.
- Cones and the spaces between them are distributed uniformly throughout the foveal array.

2. [2] Although not always shown on electromagnetic spectrum diagrams, alternating current (AC) is part of the electromagnetic spectrum. Commercial AC in the United States operates at a frequency of **60 Hz**. Determine the **wavelength in kilometres** of this component of the spectrum.

3. [3] You are tasked with designing the illumination and imaging front-end of a system that must study the shapes of cells, bacteria, viruses, and proteins. The diameters of circles required to fully enclose individual specimens in each category are **50 mm**, **1 mm**, **0.1 mm**, and **0.01 mm**, respectively. The smallest resolvable feature must be **0.001 mm**.

(a) Can this problem be solved with a *single* sensor and camera? If yes, specify the required illumination wavelength band and the type of camera.

(b) If your answer to (a) is no, propose the minimum set of illumination sources and imaging sensors required to solve the problem. For each source/sensor pair specify

[1] R.C. Gonzalez & R.E. Woods, *Digital Image Processing*, 4th ed. (Pearson, 2018), Problem 2.2.
[2] Gonzalez & Woods, Problem 2.3.
[3] Gonzalez & Woods, Problem 2.4.

the wavelength band and sensor type. (*Hint: The illumination wavelength must be no larger than the object to be imaged.*)

4. [4] A common measure of digital transmission speed is the *baud rate*, defined as bits per second. As a minimum, data are transmitted in packets comprising a start bit, one byte (8 bits) of information, and a stop bit.

(a) How many seconds would it take to transmit **500 images**, each of size 1024×1024 pixels with **256 intensity levels**, over a **3 M-baud** (10^6 bits/s) modem?
(b) Repeat for a **30 G-baud** (10^9 bits/s) modem.

5. [5] High-Definition Television (HDTV) produces interlaced images with **1125 horizontal scan lines** (two fields per frame, each field lasting $\frac{1}{60}$ s). The display aspect ratio is **16:9**. A company extracts digital colour images from HDTV video such that the number of pixels per horizontal line equals the number of scan lines multiplied by the aspect ratio. Each pixel is encoded with **24 bits** (8 bits per colour channel: R, G, B). Calculate the total number of bits required to store the images extracted from a **two-hour HDTV movie**.

6. [6] With reference to the arithmetic operations defined on digital images, prove or disprove linearity for each of the following operations:

(a) Show that **image summation** (pixel-wise addition) is a *linear* operation.
(b) Show that **image subtraction** is a *linear* operation.
(c) Show that **image multiplication** is a *nonlinear* operation.
(d) Show that **image division** is a *nonlinear* operation.

Part B Section Problems — Image Interpolation

For Problems 12–14, consider the following 2×2 source image matrix:

$$A = \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}$$

7. (Nearest Neighbour Interpolation — Square Upsampling)
Apply **Nearest Neighbour Interpolation** to upsample matrix A from 2×2 to a 3×3 image.

(a) Compute the mapping step size and determine the source coordinate corresponding to each target index using:

$$\text{Source Coordinate} = \text{Target Index} \times \frac{\text{Source Size} - 1}{\text{Target Size} - 1}$$

[4] Gonzalez & Woods, Problem 2.9.
[5] Gonzalez & Woods, Problem 2.10.
[6] Gonzalez & Woods, Problem 2.23.

(b) Apply the nearest neighbour rounding rule to each fractional source coordinate, and state which source pixel is selected for each target position.
(c) Construct and present the resulting 3×3 output matrix.
(d) Identify and explain the visual artefact produced by this method.

8. (Bilinear Interpolation — Square Upsampling)
Apply **Bilinear Interpolation** to upsample matrix A from 2×2 to a 3×3 image, using the bilinear interpolation.

(a) Determine the step size and compute the mapped source coordinates for each target index.
(b) Perform horizontal linear interpolation row by row, then apply vertical interpolation to fill all intermediate pixel values.
(c) Construct and present the resulting 3×3 output matrix.
(d) Compare this result to that of Problem 12, and discuss the quality trade-off between the two methods.

9. (Bilinear Interpolation — Non-Square Upsampling)
Apply **Bilinear Interpolation** to upsample matrix A from 2×2 to a **non-square** 3×4 matrix.

(a) Compute *independent* step sizes for the horizontal and vertical directions:

$$\text{Step}_x = \frac{\text{Source Cols} - 1}{\text{Target Cols} - 1}, \qquad \text{Step}_y = \frac{\text{Source Rows} - 1}{\text{Target Rows} - 1}$$

and list the mapped source coordinates in both dimensions.
(b) Compute each pixel value in the 3×4 output using the linear interpolation formula:

$$f(x,\, y_{\text{fixed}}) = f(x_1) + \delta\left(f(x_2) - f(x_1)\right)$$

where δ is the fractional horizontal distance from x_1.
(c) Construct and present the resulting 3×4 output matrix.
(d) Explain why separate horizontal and vertical step sizes are *necessary* when the target image dimensions differ from the source dimensions asymmetrically.